Markus Gruber, M.Sc.

AI & Automation Engineer – M&A Automation

Python · *LLM APIs · Financial Data Pipelines · Agent Workflows · RAG*

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Professional Profile



Pragmatic Senior C++/Python Backend & Systems Engineer with 9+ years of experience building real-time pipelines, cloud microservices, and high-throughput systems. Holds an M.Sc. in Scientific Computing with a strong mathematical foundation.

Resilient problem solver resolving critical production bottlenecks and memory leaks. Acts as an effective interface between development, QA, and architecture while leveraging modern AI tooling (Cursor, Copilot).

Engineering Profile

TECHNICAL FOUNDATION & SYSTEM DOMAINS

Core Systems Engineering	Platform & Architecture	Data & High-Throughput
Modern C++, C, Python, Bash	AWS (S3, RDS, EC2) & Azure	Distributed Data Pipelines & APIs
Low-Level Memory & Refactoring	Linux Ecosystem & Docker	High-Performance Query Engines
Multithreading & Deterministic Systems	Technical Governance & Code Reviews	RAG Infrastructure & Vector DBs

CORE TECHNICAL SPECIALIZATION – SYSTEMS ENGINEERING

- **High-Performance Backend Architecture:** Modern C++ system design, cache-friendly data structures, and low-latency execution in resource-critical backends.
- **System Profiling & Memory Debugging:** Root-cause analysis of memory leaks, race conditions, and latency spikes to maximize system uptime and runtime efficiency.
- **Data Pipelines & Integration Infrastructure:** Architecting high-throughput data processing and integration pipelines while automating performance verification for faster release cycles.

Professional Experience

Mindbreeze GmbH – Software / Quality Engineer (C++/Python) 09/2022 – 07/2026

C++, Python, REST APIs, Linux, Docker, Azure DevOps, Telemetry & Performance Profiling | Team: 6 Engineers

- Engineered Python/Bash automation frameworks, reducing release verification duration from 10 working days down to 3 working days.
- Optimized performance-critical backend components for enterprise architectures handling distributed data processing.
- Refactored search engine core modules and optimized index data pipelines to improve high-throughput query execution.
- Primary technical troubleshooter during release phases, resolving memory leaks and performance degradations in live systems.

Untis GmbH – Software Engineer (C++) 02/2022 – 06/2022

C++, System Refactoring, Performance Profiling, Core Desktop Modules | Team: 10 Engineers

- Refactored legacy C++/MFC core modules using Visual Studio to eliminate technical debt, optimize memory allocation, and resolve production crashes.

logi.cals – Embedded Software Engineer (C/C++) 02/2021 – 01/2022

C, C++, Embedded Systems, Safety-Critical Architectures, Deterministic Execution | Team: 6 Engineers

- Engineered deterministic C/C++ runtime execution systems (Software PLCs) and automated verification test suites for safety-standard certification across hardware platforms.

EYYES – Software Engineer (C++/Python) 04/2017 – 12/2020
C++, Python, Real-Time Execution Pipelines, CUDA, Embedded GPU Edge | Team: 8 Engineers

- Developed computer vision and deep learning pipelines (Keras, ONNX) for industrial and embedded camera systems to ensure continuous real-time edge performance.
- Engineered low-latency C++ and CUDA execution routines for real-time video stream processing on embedded GPU edge devices.
- Implemented hardware-near software components in C++ to control camera sensors, stepper motors, and SerDes interfaces.
- Optimized detection accuracy and recognition precision by training multiple TensorFlow classifiers to 99.99% accuracy through rigorous hardware-near algorithm tuning and automated benchmarking suites.

Selected Software Projects

- **Magnum C++ Game of Life:** High-performance C++ simulation focusing on memory layout and OO design (GitHub).
  Tech: Modern C++, Memory Layout, Optimization, Grid Logic
- **AWS Cloud Deep Learning Pipeline:** MLOps pipeline with automated ingestion and PyTorch inference (GitHub).
  Tech: AWS, Python, PyTorch, MLOps
- **Azure AI Document Analyzer & RAG Infra:** RAG processing backend deployed on Azure with vector databases (GitHub).
  Tech: Python, RAG, Vector DBs, REST APIs

Technical Skills

Languages & Core: Modern C++, C, Python, SQL, Java, Bash, Low-Level Memory Management, Parallel Algorithms, Scientific Computing.
Architecture & Quality: Code Reviews, Technical Governance, Clean Code, CI/CD, Microservices, REST APIs, Performance Profiling.
Cloud, Systems & Infra: Amazon AWS (S3, RDS, EC2), Microsoft Azure, Linux (Ubuntu/Debian), Windows, Docker, Kubernetes/OpenShift Basics.
Data & ML Infrastructure: RAG Infrastructure, LangChain, PySpark, DuckDB, PostgreSQL, SQLAlchemy, CUDA, OpenCL.

Education

M.Sc. Scientific Computing – University of Vienna 10/2013 – 06/2017
Thesis: Silent Data Corruption in FT-GMRES (Link, Advisor: Prof. Dr. W. Gansterer).

B.Sc. Industrial Simulation – FH St. Pölten 09/2010 – 06/2013
Thesis: Simulation of Nanocomposite Materials (Link, Advisor: Prof. Dr. T. Schrefl).

Engineering Approach & Mindset

- **Systematic Problem Solving:** First-principles analytical approach to systematic debugging and stability optimization.
- **Quality & Efficiency Focus:** Pragmatic engineering mindset balancing rapid deployment with long-term maintainability and high performance.
- **Cross-Functional Integration:** Acts as the technical glue bridging development, QA, and architecture.

Languages & Work Status

German: Native (C2) · **English:** Full Professional Proficiency (C1/C2) · **Work Permit:** EU Citizen / Austria